MAD345 - Mining II

Prospecting, Exploration & Site Investigations

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October 8, 2020

EXPLORATION ELEMENTS

GEOLOGY

Knowledge about soil, rocks, minerals, petroleum, and water

GEOPHYSICS

Measuring methods from the air, on the ground, underwater in mines, down the drillholes and in laboratory

SOIL & ROCK MECHANICS

Sampling & testing in order to establish geomechanical properties

DRILLING

Core recovery and drillhole survey

- Assessment of deposit size, its economic value, and its exploitation possibilities
- Detailed information about sub-surface, and structures for designing mines

Any mineral deposit can be successfully found and exploited only through the utilization of the full geologic principles. Compilation and interpretation of data at any stage of exploration to direct the exploration program further are the key tasks that are included in the geological studies. Aerial photography, topography map, information collected through the direct methods of prospecting, and the one through the indirect methods (geophysical and geochemical) are used for the geological interpretation, and to identify the target area.

- Area/location of the deposits and its shape
- Depth of deposits, dip and strike directions
- Thickness
- Type of surrounding rocks i.e. as overburden, hangingwall & footwall
- Grade, mineralogical and chemical composition
- Quantity and variation with respect to depth

Phases

PHASE 1: REGIONAL APPRAISAL

- Field inspection
- Structural analysis

PHASE 2: DETAILED RECONNAISSANCE

- Geologic mapping of outcrops
- Stream sediments geochemical surveys
- Magnetic surveys
- Gravity surveys

PHASE 3: DETAILED SURFACE INVESTIGATION

- Structural mapping
- Petrography, mineralogy, and trace minerals

PHASE 4: DETAILED 3D SAMPLING

- Drilling & Logging
- Mineralogical, chemical, and physical testing
- Geophysical surveys
- Resource estimation
- Preliminary valuation
- Mineral processing bulk tests

Phases

Ranking Geological and Geotechnical factors influencing key issues. 1 – Critical Influence; 2 – Major Influence; 3 – Minor Influence

PHASES & KEY ISSUES ▶	Planning				Design			Construction						
GEOLOGICAL/ GEOTECHNICAL FACTORS	Routing	Alignment	Portal Site	Tunnel Size	Excavation Stability	Tunnel Shape	Lining	Tunneling Method	Rate of Advance	Material Handling	Water Inflow	Water Pressure	Lining	Blasting
Stratigraphic/Structural														
Stratigraphic Sequence	2	2	2	3	1	1	1	1	1	2	2	3	3	1
Lithology	2	1	2	3	1	1	1	1	1	1	1	1	1	1
Folding	2	2	2	3	2	3	3	1	2	3	2	3	3	2
Faulting	1	1	1	2	1	2	1	1	1	2	1	1	1	1
Tectonic														
Seismicity	3	3	3	3	2	3	3	3	3	3	3	3	3	3
Crustal Instability	3	3	3	3	2	3	3	3	3	3	3	3	3	3
Capable Faulting	1	1	1	3	1	3	1	1	2	3	3	3	3	3
In situ Stresses	3	3	3	1	1	1	1	1	1	3	3	1	1	2
Volcanic Activity	2	2	2	3	2	3	3	3	3	3	3	3	3	3
Mechanical														
Rock Mass Strength	3	3	3	1	1	1	1	1	1	1	3	3	1	1
Deformation Moult	3	3	3	2	1	1	1	3	3	3	3	3	1	2
Discontinuities	3	3	3	2	1	1	1	2	2	2	1	1	1	1
Hydrological														
Groundwater Regime	2	2	2	3	2	3	2	1	1	2	1	1	1	2
Miscellaneous														
Mass wasting	1	1	1	3	2	3	3	3	3	3	3	3	3	3
Avalanches	1	1	1	3	2	3	3	3	3	3	3	3	3	3
Karatification	1	1	2	3	1	3	3	3	3	3	1	1	2	3
Gas	1	1	2	3	3	3	3	3	3	3	3	3	3	3

Source: www.smenet.org

Resource and Reserve Estimation

- Ore: A mineralised deposit whose characteristics have been examined and found to be **commercially viable**. The extents of the ore body are determined by the cut-off grade.

A Mineral Resource is a concentration or occurrence of diamonds, natural solid inorganic material, or natural solid fossilized organic material including base and precious metals, coal, and industrial minerals in or on the Earth's crust in such form and quantity and of such a grade or quality that it has reasonable prospects for economic extraction. The location, quantity, grade, geological characteristics and continuity of a Mineral Resource are known, estimated or interpreted from specific geological evidence and knowledge.

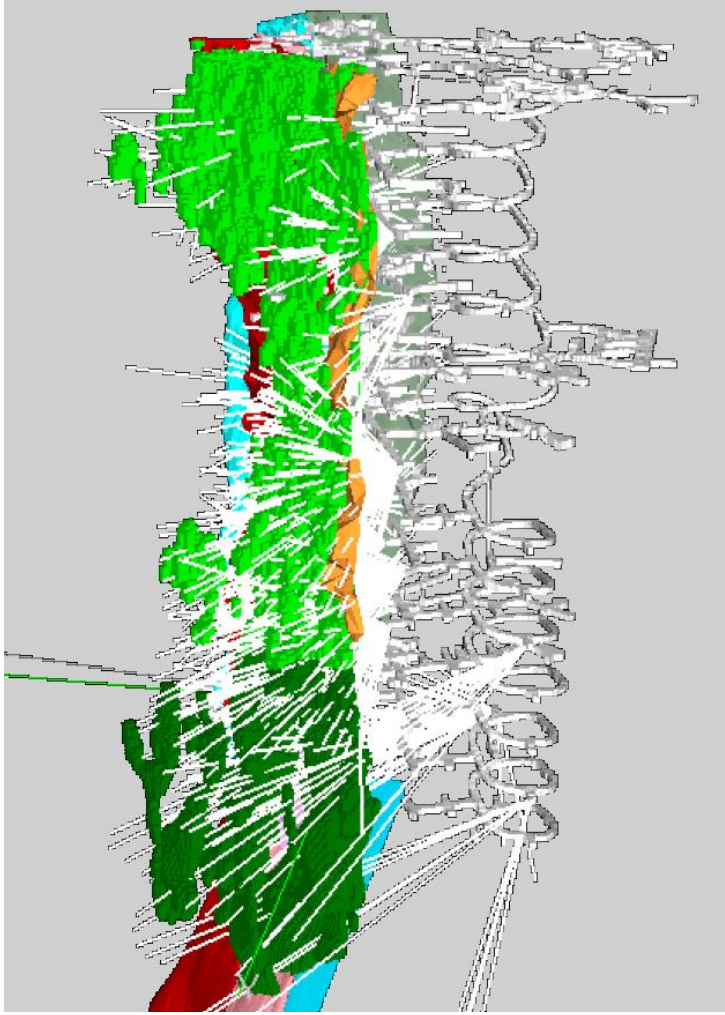
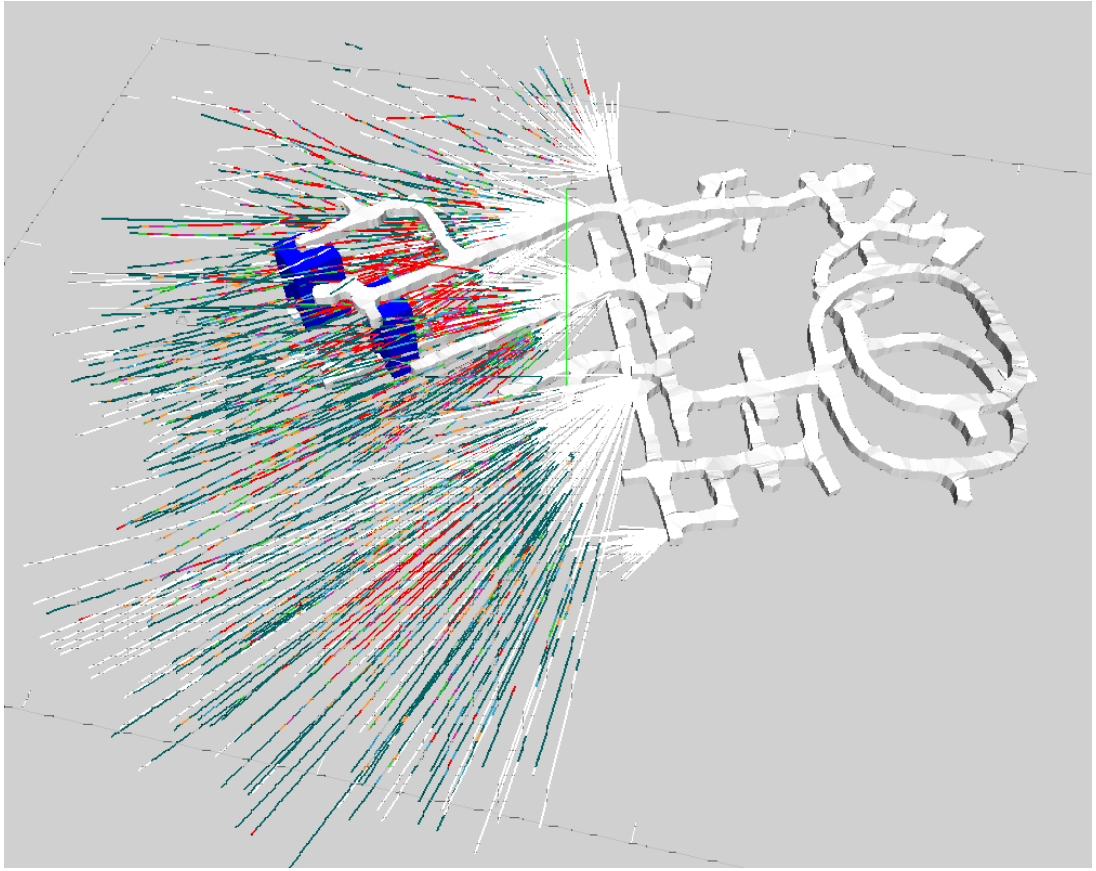
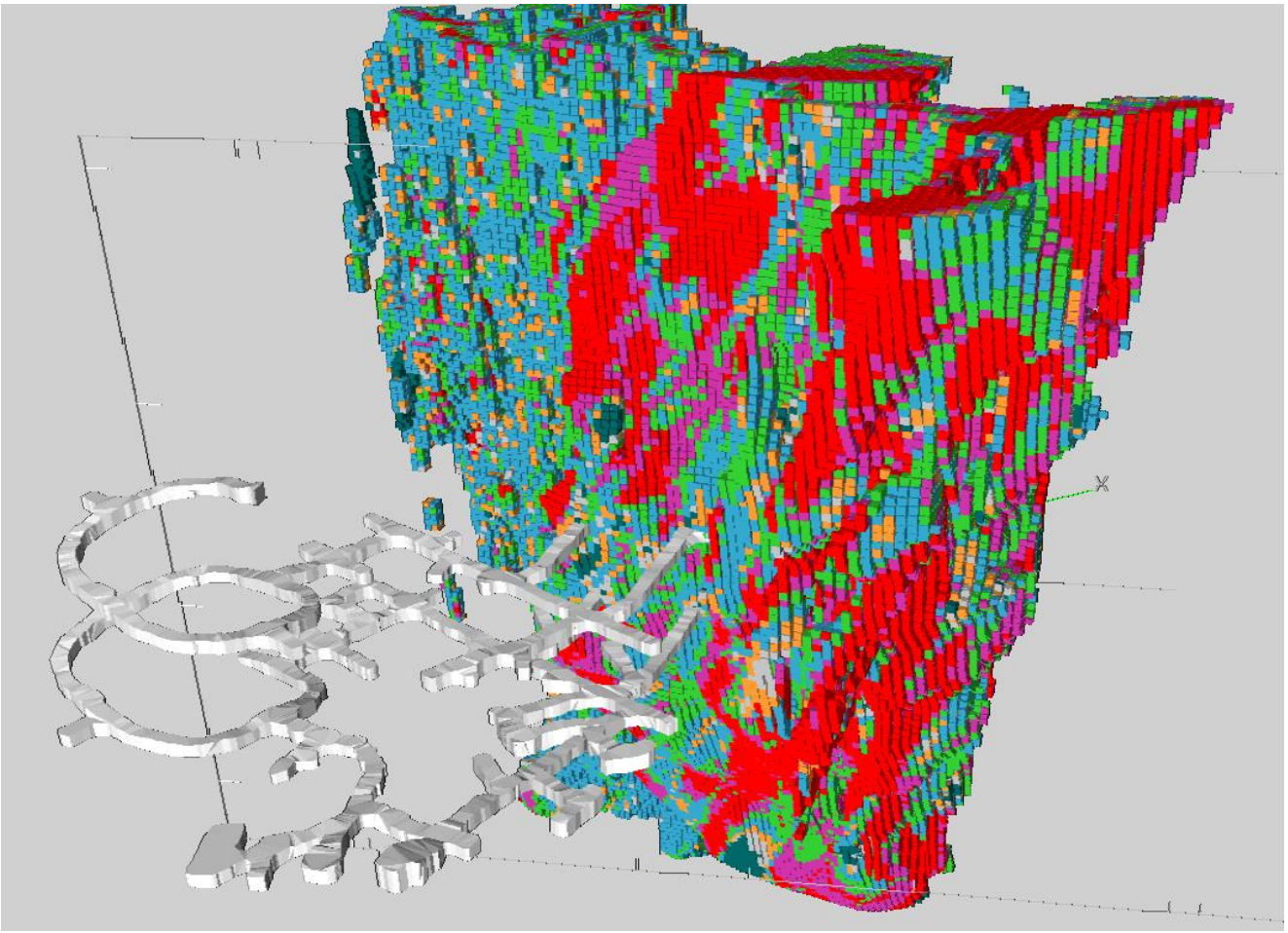
Resource and Reserve Estimation

- Ore: A mineralised deposit whose characteristics have been examined and found to be **commercially viable**. The extents of the ore body are determined by the cut-off grade.

A Mineral Reserve is the economically mineable part of a Measured or Indicated Mineral Resource demonstrated by at least a Preliminary Feasibility Study. This Study must include adequate information on mining, processing, metallurgical, economic and other relevant factors that demonstrate, at the time of reporting, that economic extraction can be justified. A Mineral Reserve includes diluting materials and allowances for losses that may occur when the material is mined.

Resource and Reserve Estimation

Qualified Person

Mineral Resource and Mineral Reserve estimates and resulting Technical Reports must be prepared by or under the direction of, and dated and signed by, a Qualified Person.

A "Qualified Person" means an individual who is an engineer or geoscientist with at least five years of experience in mineral exploration, mine development or operation or mineral project assessment, or any combination of these; has experience relevant to the subject matter of the mineral project and the technical report; and is a member or licensee in good standing of a professional association.